



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

LinkedIn Post Generator and AI News Aggregator

The domain of the Project:

Generative AI
Automation

Natural Language Processing

Team Mentors (and their designation):

Prof. Radha Kumari (Instructor)
Mr. Prujith Radhakrishanan(Trainer)

Team Member:

Mr. Shubham Yadav

Period of the project:

July 2025 to January 2026



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Declaration

The project titled “LinkedIn Post Generator and AI News Aggregator” has been mentored by Prujith Radhakrishnan, organised by SURE Trust, from July 2025 to January 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Shubham Yadav

Student, SURE Trust

Prujith Radhakrishnan

Mentor— SURE Trust

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

This project delivers a two-part solution that helps professionals **create LinkedIn-ready posts** and **stay updated with global news**:

1. **LinkedIn Post Generator** — Inspired by an end-to-end GenAI build using **Llama 3.2**, **LangChain**, **Streamlit**, and **Groq** (fast inference), the module converts user inputs/topics into **engaging, structured LinkedIn posts** with hooks, body, CTAs, emojis, and hashtags. It follows a pipeline of prompt-templating, few-shot guidance, and UI-based generation.
2. **AI News Aggregator** — A Python application that **fetches articles via RSS**, **summarizes them using OpenAI**, and **emails a formatted daily digest**. It uses feedparser for collection, OpenAI chat completions for 5–6 sentence summaries, and smtplib for distribution. The digest supports **multiple recipients**, **HTML formatting**, and can be **scheduled daily**.

Together, the system reduces manual writing time, maintains consistent professional presence, and streamlines staying informed.



Introduction

Background & Context

Maintaining a consistent presence on LinkedIn is valuable for personal branding and job opportunities, yet drafting quality posts is time-consuming. At the same time, information overload makes it hard to track daily developments across trustworthy sources. A combined tool that **generates LinkedIn-style posts** and **curates summarized news** addresses both pain points.

Problem Statement / Goals

- Professionals spend considerable time generating high-quality LinkedIn content.
- Following daily news across global outlets is tedious.
- There is a need for a **unified AI workflow** that creates posts and sends **concise news summaries** directly to email.

Scope & Limitations

Scope:

- Post generation using LLMs with configurable tone and structure.
- Aggregation from popular RSS feeds, AI summarization, and HTML email delivery.

Limitations:

- Summaries and generation quality depend on the chosen model and prompts.
- RSS coverage is limited to public feeds and may omit paywalled content.
- LinkedIn API publishing is **not** included (copy-paste/manual posting).

Innovation Component

- “**News → Summary → Email**” automated pipeline.
- “**Topic → LLM → LinkedIn style**” post generator with few-shot prompting and UI.



Project Objectives

- Build a LinkedIn post generator with **tone**, **length**, and **structure** controls.
- Implement an RSS-based news collector and **OpenAI summarizer**.
- Compose an **HTML email digest** and deliver it to one or more recipients.
- (Optional) Schedule the daily digest at a fixed time.
- Provide clear documentation, screenshots, and a GitHub repository.

Expected Outcomes & Deliverables

- Working **Streamlit/CLI** interface for post generation.
- **Daily news email** with 10–15 concise items.
- Source code, README, and usage instructions.



Methodology and Results

Technologies Used

- **Python 3.13** (core)
- **LinkedIn Post Generator:** Llama 3.2 (open-source LLM), LangChain, Groq (inference), Streamlit (UI).
- **AI News Aggregator:** feedparser, openai, schedule, python-dotenv, smtplib, email

Tools/Software

- VS Code, Git/GitHub, Browser DevTools, Gmail SMTP (for testing)

A) LinkedIn Post Generator -- Approach

1. **Prompt Design & Few-Shot Examples:** Curated exemplars that shape the voice (Professional/Motivational/Educational), structure (paragraphs/bullets/numbered), and inclusion of hooks/hashtags/CTAs.
2. **LLM Orchestration (LangChain):** Prompt templating, parameter injection (topic, tone, length), and response parsing.
3. **Fast Inference via Groq:** To reduce latency during interactive generation.
4. **Streamlit UI:** Clean inputs (topic, tone, structure, length) and real-time output for quick copy to LinkedIn.

Result: Generates on-brand, LinkedIn-ready drafts in seconds with emojis and engagement prompts; users can iterate and post with minimal editing.

B) AI News Aggregator -- Approach

Core Pipeline

1. **Environment Setup** – Loads OPENAI_API_KEY, EMAIL_USER, EMAIL_PASS, EMAIL_TO from .env using load_dotenv().
2. **Collect News via RSS** – fetch_rss_feeds() iterates through curated feeds (CNN, NYT, BBC, Al Jazeera, The Guardian, TechCrunch, r/worldnews) and extracts title, link, summary.
3. **Summarize with OpenAI** – summarize_text() calls client.chat.completions.create() with model="gpt-4o-mini" to compress each item into **5–6 sentences** (limited to max_tokens=80).
4. **Build HTML Digest** – build_html_email() generates an HTML layout with **headline (clickable link)** and **summary** for each item.
5. **Send Email** – send_email() composes a multipart message and delivers via Gmail SMTP (TLS, port 587). Your final version supports **multiple recipients** (comma-separated).



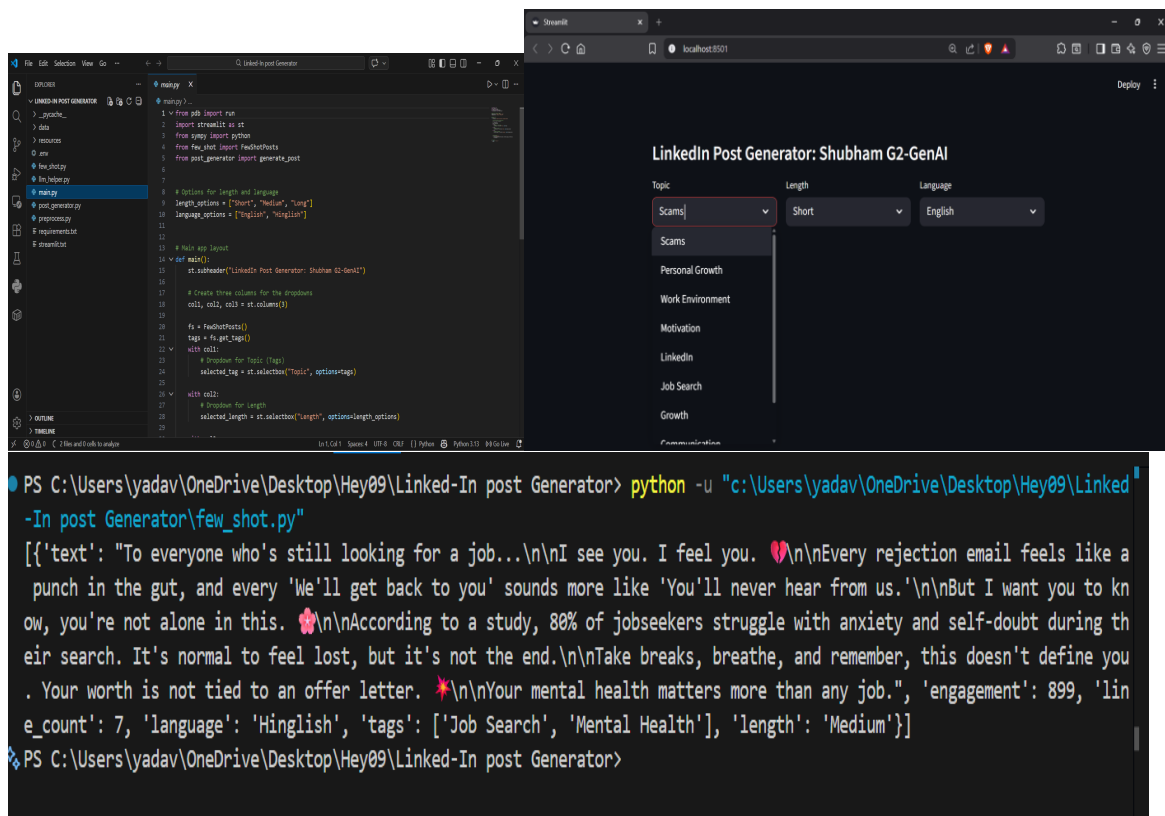
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Project Architecture (Clearly Describe the Architecture)

High-Level Diagram (Textual)

- LinkedIn Post Generator
 - UI (Streamlit) → Prompt Builder & Few-Shot Templates (LangChain) → LLM (Llama 3.2 via Groq) → Post Formatter (hooks, emojis, hashtags) → User Copies to LinkedIn.
- AI News Aggregator
 - RSS Sources → Fetcher (feedparser) → Summarizer (OpenAI Chat Completions) → HTML Composer → Email Sender (SMTP) → (Optional) Scheduler schedule.

The YouTube walkthrough (Codebasics) directly influenced the LLM + LangChain + Streamlit flow and few-shot methodology for LinkedIn-style generation.





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Streamlit

localhost:8501

Deploy

LinkedIn Post Generator: Shubham G2-GenAI

Topic

Length

Language

LinkedIn

Long

Hinglish

Generate

"Unlocking the Power of LinkedIn for Professional Growth

As a professional, you know how crucial it is to stay connected with your network and showcase your skills. LinkedIn is a powerful platform that can help you achieve just that.

With over 900 million users worldwide, LinkedIn is the ideal place to establish yourself as a thought leader in your industry.

Whether you're a fresher or an experienced professional, LinkedIn offers a range of features to help you grow your career.

```
File Edit Selection View Go ... Linked-In post Generator
```

EXPLORER

LINKED-IN POST GENERATOR

main.py

```
1 from pdb import run
2 import streamlit as st
3 from sympy import python
4 from few_shot import FewShotPosts
5 from post_generator import generate_post
6
7
8 # Options for length and language
9 length_options = ["Short", "Medium", "Long"]
10 language_options = ["English", "Hinglish"]
11
12
13 # Main app layout
14 def main():
15     st.subheader("LinkedIn Post Generator: Shubham G2-GenAI")
16
17     # Create three columns for the dropdowns
18     col1, col2, col3 = st.columns(3)
19
20     fs = FewShotPosts()
21     tags = fs.get_tags()
22     with col1:
23         # Dropdown for Topic (Tags)
24         selected_tag = st.selectbox("Topic", options=tags)
25
26     with col2:
27         # Dropdown for Length
28         selected_length = st.selectbox("Length", options=length_options)
29
30     # Generate post
31     post = generate_post(selected_tag, selected_length)
```

Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Python Python 3.13 Go Live



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The screenshot displays a Jupyter Notebook environment with the following code and output:

```
ai_news_aggregator_mail.ipynb > pip install feedparser openai schedule python-dotenv
Generate + Code + Markdown Run All Clear All Outputs Outline ... Python 3.13.12

print("✅ Daily News Email Sent Successfully")

[10]

# STEP 6: SCHEDULER

daily_news_job()
# schedule.every().day.at("08:00").do(daily_news_job)

# print("🔴 Scheduler started...")

# while True:
#     schedule.run_pending()
#     time.sleep(30)

print("✅ Job completed. Exiting program.")
print("✅ Test run finished. Exiting.")

[11]

...
Running Daily News Pipeline...
✅ Daily News Email Sent Successfully
✅ Job completed. Exiting program.
✅ Test run finished. Exiting.
```

The Outlook email interface shows a message from shubhamyadav2genai@gmail.com titled "AI-POWERED Daily News Digest". The email content includes the following sections:

AI-POWERED DAILY NEWS AGGREGATOR

[Carney Speech on U.S. 'Rupture' and Canada's Survival Draws Standing Ovation at Davos](#)

At the World Economic Forum in Davos, Prime Minister Mark Carney received a standing ovation for his bold remarks on the waning influence of Pax Americana. He emphasized the need for his country to adapt to a shifting global landscape and is actively seeking new allies to strengthen their position. Carney's comments reflect concerns over declining American leadership on the world stage and the implications it has for international cooperation.

[Canada's Leader Warns of a 'Rupture' in the Global Order, and ICE Is Accused of Killing Detainee](#)

The article discusses the declining emphasis on providing education in schools, raising questions about its

GitHub - [Link](#)

if any problem occurs refer to - <https://github.com/sure-trust/SHUBHAM-YADAV-g2-genai>



Learning and Reflection

- **New Learnings (Technology & Management)**
- **LLM Prompting & Few-Shot Design:** Crafting templates that reliably produce LinkedIn-style content; tuning tone/structure and adding engagement CTAs.
- **LangChain Orchestration:** Managing prompts and parameters for consistent outputs.
- **Rapid Inference with Groq:** Understanding latency trade-offs for interactive UX.
- **Web Data Feeds (RSS):** Using feedparser to normalize multi-source content.
- **Email Automation:** Building compliant HTML for mail clients and handling multiple recipients.
- **Project Planning:** Dividing work into modules, version control on GitHub, and test runs.
- **Team Experience (Overall)**
- The team experienced the complete lifecycle: requirement scoping, architecture design, iterative prototyping, prompt tuning, error handling, and user-facing polish (UI, email formatting). The hands-on exposure to both **generative** and **extractive** AI tasks was invaluable.



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Conclusion and Future Scope

Recap of Objectives & Achievements

- Built a **LinkedIn Post Generator** using an industry-aligned stack (Llama 3.2 + LangChain + Streamlit + Groq) that outputs structured, platform-ready posts.
- Implemented an **AI News Aggregator** that fetches, summarizes, and emails a daily digest from multiple sources.
- Documented code, produced screenshots, and prepared for GitHub publication.

Future Scope

- **Direct LinkedIn API posting** and scheduled posting (subject to API access).
- **Topic targeting** (filters: AI only, region, company tags).
- **Multilingual generation** for broader reach.
- **Entity-aware summaries** (NER to highlight people/companies).
- **Serverless deployment** (Cloud Run/Lambda) and background scheduling.
- **Replace SMTP with email APIs** (SendGrid, SES) for reliability.